

ADITI KUMARI

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SUMMARY

Final-year Computer Science Engineering student skilled in Machine Learning, Data Analytics, and Backend Development with hands-on experience in Python, SQL, Scikit-learn, Flask, Spring Boot, REST APIs, and RAG-based AI systems. Research Intern at ISRO with experience in real-time predictive modeling and scalable analytical applications. Seeking opportunities in AI/ML Engineering, Data Engineering, and Software Development.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Machine Learning & AI: Scikit-learn, NumPy, Feature Engineering, Machine Learning Pipelines, Vector Embeddings

Generative AI: Retrieval-Augmented Generation (RAG), FAISS, Prompt Engineering, Retrieval Patterns

Backend Development: Spring Boot, Flask, REST APIs, MVC Architecture, Authentication and Authorization

Databases & Cloud: PostgreSQL, MongoDB, Supabase, AWS, SQL Databases

Data Engineering & Analytics: Data Preprocessing, ETL Concepts, Data Visualization, Data Modeling

Tools & Technologies: Git, GitHub, Docker, Postman, Selenium, Linux, API Testing

Core CS Fundamentals: Data Structures & Algorithms, OOPs, DBMS, Operating Systems

Soft Skills: Analytical Problem Solving, Team Collaboration, Research & Analysis, Agile SDLC, Adaptability

WORK EXPERIENCE

Research Intern | Indian Space Research Organisation (ISRO) | Ahmedabad, India Jul 2025- Sep 2025

- Developed algorithms for real-time Low Earth Orbit (LEO) satellite tracking and trajectory prediction using TLE data.
- Reduced dependency on costly satellite telemetry data by implementing TLE-based prediction models.
- Built predictive models for satellite position and velocity estimation with ~80% accuracy using orbital mechanics.
- Collaborated with researchers and scientists on analytical models for space situational awareness applications.

Data Science Intern | ElevenceSkills | Tamil Nadu, India Dec 2025 -May 2026

- Built a RAG chatbot using Python, FAISS, and vector embeddings, improving response relevance by ~35%.
- Automated vector database updates, reducing manual effort by ~70%.
- Reduced LLM hallucinations by ~40% through optimized contextual retrieval.

EDUCATION

B.Tech (Computer Science Engineering) | Savitribai Phule Pune University (SPPU) | CGPA: 8.18 | Aug 2022-2026

HSC | Kendriya Vidyalaya, Mumbai | CGPA: 7.90 | Apr 2020-2021

SSC | Kendriya Vidyalaya, Mumbai | CGPA: 7.40 | Apr 2018-2019

PROJECTS

LEO Satellite Tracking and Collision Alert System using TLE Data| Python, Flask, CesiumJS | Apr 2025 -Jul 2025

- Developed a real-time satellite monitoring system using TLE data and interactive 3D visualization.
- Implemented RK4 and J2 perturbation models for orbit prediction and conjunction analysis.
- Integrated authentication systems and analytical dashboards for secure monitoring.
- Achieved ~88% short-term orbit prediction accuracy through optimized propagation algorithms.

Online Food Ordering System |Spring Boot, MongoDB, Spring Security | Aug 2025 -Nov 2025

- Developed a scalable full-stack food ordering platform with role-based authentication supporting 3 user roles.
- Built 15+ RESTful APIs using Spring Boot and MVC architecture, improving backend modularity and scalability.
- Implemented a 5-stage real-time order workflow, reducing order processing delays by ~30%.
- Optimized MongoDB schema design with 3 collections, improving data retrieval efficiency and API performance.

RasoiBot - AI-Powered Recipe Recommendation System | React ,KNN, NumPy, OpenCV | Jan 2025 - Apr 2025

- Built an AI-powered recipe recommendation system using KNN and cosine similarity algorithms.
- Developed ingredient-based recommendation and food image recognition features using OpenCV.
- Achieved ~94% recommendation accuracy through optimized feature engineering techniques.

CERTIFICATIONS

AWS Educate Introduction to Cloud 101 | Amazon Web Services Training and Certification Apr 2026

AWS Educate Introduction to Generative AI | Amazon Web Services Training and Certification Mar 2026

Introduction to Data Engineering | Coursera Apr 2025

ACHIEVEMENTS

- Published a research paper on TLE-based LEO satellite tracking and collision prediction systems.
- Solved 100+ DSA problems on LeetCode focusing on algorithms and problem-solving.
- Earned NPTEL certification in Internet of Things (IoT) with 85% score (May 2026).